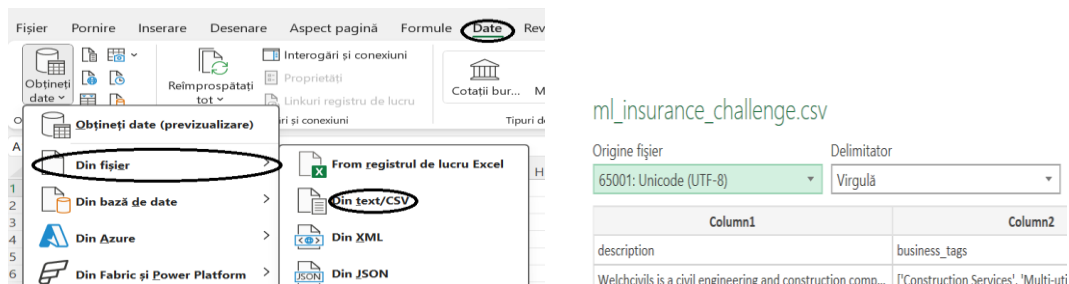


Company Classifier

Pentru început am analizat fișierul **"ml_insurance_challenge.csv"** și am observat că este voluminos și greu de curățat.

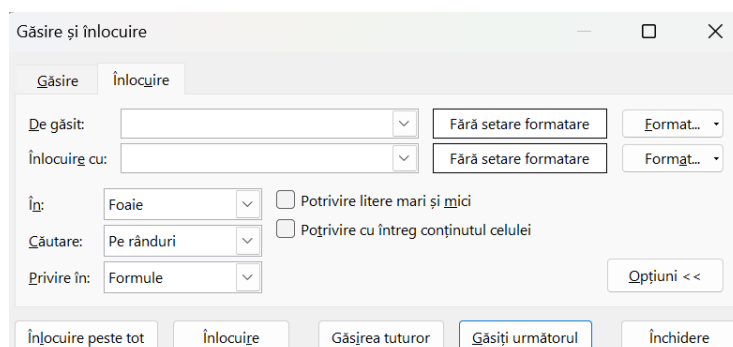
Am considerat ca este nevoie să îl curăț, pentru a ușura munca din spatele codului. Astfel, am folosit câteva funcții în Excel, cel puțin pentru o parte din „probleme”.

La început, am deschis o filă nouă de Excel pentru a transforma într-un tabel citibil.



Tabelul original	Tabelul obținut																																																																																																																																																																																																																																																																																																																																										
<table><tr><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th><th>G</th><th>H</th><th>I</th><th>J</th><th>K</th></tr><tr><td>description</td><td>business</td><td>sector</td><td>category</td><td>niche</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>WelchCivils is a civil engineering and construction company that specializes in designing and building utility network connections across the UK. They offer multi-utility solutions that combine electricity, gas, water</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Kyoto Vegetable Specialists (Ukemo, also known as Iwa-machi), is a company based in Kyoto, Japan that specializes in the sale of vegetables. They have been in business for ten years and offer a collection of veget</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Loidholdho Integrative Holdingsenscraft is a company that offers a range of services and products to its customers. Their products are all handmade and of the highest quality, produced on a biodynamic basis wit</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>PATAGONIA Chicago PTintura is an auto body shop located in Comodoro Rivadavia, Chubut Province, Argentina. The company specializes in providing auto body repair services, ["Automotive Body Repair Services",</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Stanica WODNA PTITN Swomyspace is a cultural establishment located in Swomyspace, Poland. It is a popular destination for kayakers and tourists of all levels, offering a variety of activities and events. The es</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>BIQ Benefit-cos is a Brazilian company that specializes in generating benefits for both individuals and businesses. With a structure of three large companies in the food sector and 20 years of experience, BIQ Bene</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>EnvCompli is a company that provides personalized Industrial General Permit (IGP) Management services for Storm Water Compliance needs. They offer an assigned specialist to meet all your needs, coordinate y</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Corrugated Beam Form Ltd is a company based in Walsall, England, specializing in the supply of concrete products, ["Cut Sheets", "Formwork Installation Services", "Corner Joints", "Lightweight Polypropylene Manu</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>The company specializes in the production and distribution of packaging materials, including cardboard sealing tape, wholesale packaging, and packaging supplies. They also offer printed packaging services, ["C</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Indira Caterers And Event Managerr is a company categorized under restaurant supply stores. It is located in Alappuzha, Kerala, India. The company specializes in providing supplies for restaurants and event manag</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Halle Heat Exchange Equipment Co., Ltd is a Chinese company that specializes in the production and sales of heat exchangers. Their main products include plate-type heat exchanger, welded panel type heat exch</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Ballymcanthy Elders Housing Association Ltd. is a registered charity in Ireland that aims to provide quality, age-friendly, and sustainable homes for older people in the local community who require them. Their goa</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>An Mhien Scientific Equipment Co., Ltd. (AnMhien) is a company that specializes in the distribution of scientific equipment, reagents, probiotics, and consumables for various industries such as life science resear</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Anovei is a company that specializes in air treatment, purifier leasing, and nasal filtration research and development, as well as production, ["Innovative Solutions Provider", "Technology Solutions", "High-tech Solu</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Do Motor Service di Damiano Barbato is an auto electrical service and auto repair shop located in Arzano, Campania, Italy, ["Diagnostic Services", "Correct Diagnosis and Maintenance", "Automobiles And Motorcy</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>DOOO B&B Th Petrowac is a Slovakian company that specializes in the production of raw materials and the wholesale of cleaning products. Their main focus is on producing raw powders and selling cleaning items, ["</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	A	B	C	D	E	F	G	H	I	J	K	description	business	sector	category	niche							WelchCivils is a civil engineering and construction company that specializes in designing and building utility network connections across the UK. They offer multi-utility solutions that combine electricity, gas, water											Kyoto Vegetable Specialists (Ukemo, also known as Iwa-machi), is a company based in Kyoto, Japan that specializes in the sale of vegetables. They have been in business for ten years and offer a collection of veget											Loidholdho Integrative Holdingsenscraft is a company that offers a range of services and products to its customers. Their products are all handmade and of the highest quality, produced on a biodynamic basis wit											PATAGONIA Chicago PTintura is an auto body shop located in Comodoro Rivadavia, Chubut Province, Argentina. The company specializes in providing auto body repair services, ["Automotive Body Repair Services",											Stanica WODNA PTITN Swomyspace is a cultural establishment located in Swomyspace, Poland. It is a popular destination for kayakers and tourists of all levels, offering a variety of activities and events. The es											BIQ Benefit-cos is a Brazilian company that specializes in generating benefits for both individuals and businesses. With a structure of three large companies in the food sector and 20 years of experience, BIQ Bene											EnvCompli is a company that provides personalized Industrial General Permit (IGP) Management services for Storm Water Compliance needs. They offer an assigned specialist to meet all your needs, coordinate y											Corrugated Beam Form Ltd is a company based in Walsall, England, specializing in the supply of concrete products, ["Cut Sheets", "Formwork Installation Services", "Corner Joints", "Lightweight Polypropylene Manu											The company specializes in the production and distribution of packaging materials, including cardboard sealing tape, wholesale packaging, and packaging supplies. They also offer printed packaging services, ["C											Indira Caterers And Event Managerr is a company categorized under restaurant supply stores. It is located in Alappuzha, Kerala, India. The company specializes in providing supplies for restaurants and event manag											Halle Heat Exchange Equipment Co., Ltd is a Chinese company that specializes in the production and sales of heat exchangers. Their main products include plate-type heat exchanger, welded panel type heat exch											Ballymcanthy Elders Housing Association Ltd. is a registered charity in Ireland that aims to provide quality, age-friendly, and sustainable homes for older people in the local community who require them. Their goa											An Mhien Scientific Equipment Co., Ltd. (AnMhien) is a company that specializes in the distribution of scientific equipment, reagents, probiotics, and consumables for various industries such as life science resear											Anovei is a company that specializes in air treatment, purifier leasing, and nasal filtration research and development, as well as production, ["Innovative Solutions Provider", "Technology Solutions", "High-tech Solu											Do Motor Service di Damiano Barbato is an auto electrical service and auto repair shop located in Arzano, Campania, Italy, ["Diagnostic Services", "Correct Diagnosis and Maintenance", "Automobiles And Motorcy											DOOO B&B Th Petrowac is a Slovakian company that specializes in the production of raw materials and the wholesale of cleaning products. Their main focus is on producing raw powders and selling cleaning items, ["											<table><tr><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th><th>G</th><th>H</th><th>I</th><th>J</th><th>K</th></tr><tr><td>1</td><td>description</td><td>business</td><td>sector</td><td>category</td><td>niche</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>2</td><td>welchcivils Constructi</td><td>Services</td><td>Civil Engin</td><td>Other Heavy and Civil Engineering</td><td>Construction</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>3</td><td>kyoto vege</td><td>Wholesale Manufactu</td><td>Fruit & Veg</td><td>Frozen Fruit, Juice, and Vegetable</td><td>Manufacturing</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>4</td><td>loldholdho Living Form Manufactu</td><td>Farms & A</td><td>All Other</td><td>Miscellaneous Crop</td><td>Farming</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>5</td><td>patagonia Automotiv Services</td><td>Auto Body</td><td>Automotive Body, Paint, and Interior</td><td>Repair and Maintenance</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>6</td><td>stanica wc Cultural Ar Services</td><td>Boat Tours</td><td>Scenic and Sightseeing</td><td>Transportation, Water</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>7</td><td>biq benefic Healthy Fo Services</td><td>Payment P</td><td>Financial Transactions</td><td>Processing, Reserve, and Clearinghouse</td><td>Activities</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>8</td><td>envcompli Source Co Services</td><td>Environme</td><td>Environmental Consulting</td><td>Services</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>9</td><td>corrugatec Cut Sheet: Manufactu</td><td>Building M</td><td>Engineered Wood Member</td><td>Manufacturing</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>10</td><td>the compa C Packagih</td><td>Wholesale Paper & C</td><td>Industrial and Personal Service</td><td>Paper Merchant</td><td>Wholesalers</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>11</td><td>indira cate Event Orgs</td><td>Wholesale Restaura</td><td>Other Commercial</td><td>Equipment Merchant</td><td>Wholesalers</td><td></td><td></td><td></td><td></td><td></td></tr></table>	A	B	C	D	E	F	G	H	I	J	K	1	description	business	sector	category	niche						2	welchcivils Constructi	Services	Civil Engin	Other Heavy and Civil Engineering	Construction						3	kyoto vege	Wholesale Manufactu	Fruit & Veg	Frozen Fruit, Juice, and Vegetable	Manufacturing						4	loldholdho Living Form Manufactu	Farms & A	All Other	Miscellaneous Crop	Farming						5	patagonia Automotiv Services	Auto Body	Automotive Body, Paint, and Interior	Repair and Maintenance							6	stanica wc Cultural Ar Services	Boat Tours	Scenic and Sightseeing	Transportation, Water							7	biq benefic Healthy Fo Services	Payment P	Financial Transactions	Processing, Reserve, and Clearinghouse	Activities						8	envcompli Source Co Services	Environme	Environmental Consulting	Services							9	corrugatec Cut Sheet: Manufactu	Building M	Engineered Wood Member	Manufacturing							10	the compa C Packagih	Wholesale Paper & C	Industrial and Personal Service	Paper Merchant	Wholesalers						11	indira cate Event Orgs	Wholesale Restaura	Other Commercial	Equipment Merchant	Wholesalers					
A	B	C	D	E	F	G	H	I	J	K																																																																																																																																																																																																																																																																																																																																	
description	business	sector	category	niche																																																																																																																																																																																																																																																																																																																																							
WelchCivils is a civil engineering and construction company that specializes in designing and building utility network connections across the UK. They offer multi-utility solutions that combine electricity, gas, water																																																																																																																																																																																																																																																																																																																																											
Kyoto Vegetable Specialists (Ukemo, also known as Iwa-machi), is a company based in Kyoto, Japan that specializes in the sale of vegetables. They have been in business for ten years and offer a collection of veget																																																																																																																																																																																																																																																																																																																																											
Loidholdho Integrative Holdingsenscraft is a company that offers a range of services and products to its customers. Their products are all handmade and of the highest quality, produced on a biodynamic basis wit																																																																																																																																																																																																																																																																																																																																											
PATAGONIA Chicago PTintura is an auto body shop located in Comodoro Rivadavia, Chubut Province, Argentina. The company specializes in providing auto body repair services, ["Automotive Body Repair Services",																																																																																																																																																																																																																																																																																																																																											
Stanica WODNA PTITN Swomyspace is a cultural establishment located in Swomyspace, Poland. It is a popular destination for kayakers and tourists of all levels, offering a variety of activities and events. The es																																																																																																																																																																																																																																																																																																																																											
BIQ Benefit-cos is a Brazilian company that specializes in generating benefits for both individuals and businesses. With a structure of three large companies in the food sector and 20 years of experience, BIQ Bene																																																																																																																																																																																																																																																																																																																																											
EnvCompli is a company that provides personalized Industrial General Permit (IGP) Management services for Storm Water Compliance needs. They offer an assigned specialist to meet all your needs, coordinate y																																																																																																																																																																																																																																																																																																																																											
Corrugated Beam Form Ltd is a company based in Walsall, England, specializing in the supply of concrete products, ["Cut Sheets", "Formwork Installation Services", "Corner Joints", "Lightweight Polypropylene Manu																																																																																																																																																																																																																																																																																																																																											
The company specializes in the production and distribution of packaging materials, including cardboard sealing tape, wholesale packaging, and packaging supplies. They also offer printed packaging services, ["C																																																																																																																																																																																																																																																																																																																																											
Indira Caterers And Event Managerr is a company categorized under restaurant supply stores. It is located in Alappuzha, Kerala, India. The company specializes in providing supplies for restaurants and event manag																																																																																																																																																																																																																																																																																																																																											
Halle Heat Exchange Equipment Co., Ltd is a Chinese company that specializes in the production and sales of heat exchangers. Their main products include plate-type heat exchanger, welded panel type heat exch																																																																																																																																																																																																																																																																																																																																											
Ballymcanthy Elders Housing Association Ltd. is a registered charity in Ireland that aims to provide quality, age-friendly, and sustainable homes for older people in the local community who require them. Their goa																																																																																																																																																																																																																																																																																																																																											
An Mhien Scientific Equipment Co., Ltd. (AnMhien) is a company that specializes in the distribution of scientific equipment, reagents, probiotics, and consumables for various industries such as life science resear																																																																																																																																																																																																																																																																																																																																											
Anovei is a company that specializes in air treatment, purifier leasing, and nasal filtration research and development, as well as production, ["Innovative Solutions Provider", "Technology Solutions", "High-tech Solu																																																																																																																																																																																																																																																																																																																																											
Do Motor Service di Damiano Barbato is an auto electrical service and auto repair shop located in Arzano, Campania, Italy, ["Diagnostic Services", "Correct Diagnosis and Maintenance", "Automobiles And Motorcy																																																																																																																																																																																																																																																																																																																																											
DOOO B&B Th Petrowac is a Slovakian company that specializes in the production of raw materials and the wholesale of cleaning products. Their main focus is on producing raw powders and selling cleaning items, ["																																																																																																																																																																																																																																																																																																																																											
A	B	C	D	E	F	G	H	I	J	K																																																																																																																																																																																																																																																																																																																																	
1	description	business	sector	category	niche																																																																																																																																																																																																																																																																																																																																						
2	welchcivils Constructi	Services	Civil Engin	Other Heavy and Civil Engineering	Construction																																																																																																																																																																																																																																																																																																																																						
3	kyoto vege	Wholesale Manufactu	Fruit & Veg	Frozen Fruit, Juice, and Vegetable	Manufacturing																																																																																																																																																																																																																																																																																																																																						
4	loldholdho Living Form Manufactu	Farms & A	All Other	Miscellaneous Crop	Farming																																																																																																																																																																																																																																																																																																																																						
5	patagonia Automotiv Services	Auto Body	Automotive Body, Paint, and Interior	Repair and Maintenance																																																																																																																																																																																																																																																																																																																																							
6	stanica wc Cultural Ar Services	Boat Tours	Scenic and Sightseeing	Transportation, Water																																																																																																																																																																																																																																																																																																																																							
7	biq benefic Healthy Fo Services	Payment P	Financial Transactions	Processing, Reserve, and Clearinghouse	Activities																																																																																																																																																																																																																																																																																																																																						
8	envcompli Source Co Services	Environme	Environmental Consulting	Services																																																																																																																																																																																																																																																																																																																																							
9	corrugatec Cut Sheet: Manufactu	Building M	Engineered Wood Member	Manufacturing																																																																																																																																																																																																																																																																																																																																							
10	the compa C Packagih	Wholesale Paper & C	Industrial and Personal Service	Paper Merchant	Wholesalers																																																																																																																																																																																																																																																																																																																																						
11	indira cate Event Orgs	Wholesale Restaura	Other Commercial	Equipment Merchant	Wholesalers																																																																																																																																																																																																																																																																																																																																						

Pe parcurs ce căutam un model potrivit de ML și descopeream ce „piedici” pot apărea, am aflat că pe lângă un tabel citibil este indicat să nu existe rânduri goale, fiindcă este necesar același număr de rânduri pentru fiecare coloană. Am eliminat din coloana „business_tags” parantezele pătrate și rândurile goale le-am notat cu NaN, folosind Crlt+A și Crlt+H. De asemenea, am eliminat duplicatele folosind aceleași funcții.



Am mai încercat și alte funcții, precum `lowercase`, pentru că erau cuvinte scrise doar cu litera mari și nu era citibil și nici estetic. Doar că, am considerat că nu e folositoare această funcție, pentru că se pierde din farmecul propoziției de a începe fraza sau un nume propriu cu o litera mică.

Am concluzionat că Excel nu m-ar ajuta să rezolv asocierea dintre cele două fișiere csv ("**ml_insurance_challenge.csv**" și "**taxonomy.csv**") și am trecut la Python, pentru că avea mai multe librării folositoare. De asemenea, folositor am citit că era și limbajul R (fiind un limbaj folosit tot pentru analiză), dar nu are atât de multe librării ca și Python. În final, chiar dacă am făcut modificările de mai sus în Excel, în codul tot fișierul ("**ml_insurance_challenge.csv**") original l-am folosit.

Aici am încercat prin cod să curăț unele probleme, precum duplicatele `date = pd.read_csv('CompanyDetails.csv', sep=';') date = date.drop_duplicates()`. Chiar am încercat să creez un grafic cu valorile lipsă, crezând că-mi va fi mai ușor să identific ce date am dacă elimin din ecuație ce era lipsă.

```
cols = ['description', 'business_tags', 'category', 'niche']  
  
missing_counts = date[cols].nunique().sum()  
  
plt.title('Număr de valori lipsă pe coloană')  
  
plt.xlabel('Coloană')  
  
plt.ylabel('Număr NaN')  
  
plt.show()
```

Dar, n-am avut succes nici așa.

Astfel, a fost nevoie să caut modelul potrivit de ML care să poată face față unui csv care are text în mai multe limbi (chineză, engleză), text foarte lung, semne de punctuație (",", "?"), caractere asiatice (đ, lý, bia, nguy?n...), simboluri (+-, ©).

Am descoperit mai multe modele și am analizat care ar fi cel potrivit pentru datele mele. Câteva dintre ele:

Model	Detalii
BERT	înțelege text în engleză
mBERT	rezultate mai bune decât BERT
XLM-R	cel mai bun multilingv pentru NLU(Natural Language Understanding)

Am ales modelul XLM-R. XLM-R este un **encoder Transformer**. Asta înseamnă că, pentru orice text, el returnează **vectori numerici** (embedding-uri) care reprezintă sensul aceluia text. Odată ce sunt obținute embedding-urile, trebuie să se măsoare **cât de apropiate**

sunt cele două texte. Astfel, este obținută **similaritatea cosinus**. Aceasta spune că dacă unghiul dintre cei doi vectori este ~ 1 înseamnă că textele sunt aproape identice ca sens sau 0 nu împărtășesc semnificație comună.

➤ Librăriile folosite sunt

```
# import pandas as pd pentru a citi și manipula datele
```

```
# import numpy as np pentru numere
```

```
# from matplotlib import pyplot as plt pentru grafice
```

```
# import seaborn as sns pentru a transforma textul în vectori (embeddings)
```

După ce am citit csv-urile `dfA = pd.read_csv('ml_insurance_challenge.csv', sep=',')` și am combinat toată informația într-un text, am încărcat modelul XLM-R.

Am transformat textul în embedding `model = SentenceTransformer("sentence-transformers/paraphrase-xlm-r-multilingual-v1")` și apoi am căutat similitudinea cosinus. Iar la final, am salvat totul într-un fișier nou `matches_df.to_csv("final.csv", index=False, encoding="utf-8-sig") print("SAVE 'final.csv')"`.